SCHOOL OF COMPUTER SCIENCE AND INFORMATION SYSTEMS

FALL SEMESTER 2025-26

Course Code: SWE2035 - Big Data Technologies

Slot : E2+TE2

Faculty: Dr. Padmakumari P

Course Project Summary

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Project Review **Date: 31/10/2025**

Faculty Signature

Title of your project:

Mental Health Monitoring and wellness Recommendation

Introduction:

- Mental health is a vital aspect of human well-being, shaping our emotions, thoughts, and behaviors in everyday life.
- With the increasing pace of modern life and the rise of technology, mental health concerns such as stress, anxiety, and depression have become more common.
- Factors like excessive workload, social disconnection, and unhealthy lifestyle habits often contribute to these problems.
- Although early diagnosis and proper care can greatly improve mental health, many individuals avoid seeking help due to social stigma, lack of knowledge, or limited access to mental health resources.
- To overcome these challenges, Mental Health Monitoring and Wellness Recommendation systems utilize advanced technologies such as artificial intelligence (AI), machine learning (ML), and wearable devices to track an individual's emotional and psychological well-being.
- By examining behavior patterns, physiological signals, and self-reported information, these systems can identify signs of mental strain and provide customized suggestions for wellness improvement.
- Such intelligent systems encourage users to take charge of their mental health and support healthcare professionals with real-time, data-based insights.
- Integrating technology with mental health care promotes a comprehensive approach to well-being—enhancing awareness, preventing mental health issues, and fostering a balanced and mentally resilient community.
- They also promote a personalized approach to self-care by recommending mindfulness activities, stress-relief exercises, and healthy lifestyle practices.
- Overall, merging digital technology with mental wellness offers an effective, accessible, and user-friendly way to support individuals in maintaining emotional balance and long-term psychological resilience.

Scope of Project:

- The project aims to design and develop an intelligent system that monitors users' mental health in real time.
- It focuses on analyzing user inputs, such as mood, sleep patterns, and activity levels, to assess emotional well-being.

- The system will use AI and ML algorithms to predict potential mental health issues like stress or anxiety.
- It will provide personalized wellness recommendations, including relaxation methods, physical exercises, and lifestyle tips.
- The platform can integrate with wearable devices and mobile applications for continuous health tracking.
- It offers users an interactive interface to record their mental state and receive feedback.
- The system can generate reports to help users and healthcare professionals monitor mental health progress over time.
- It aims to raise awareness about mental health and promote preventive self-care habits.
- The project ensures data privacy and confidentiality while handling sensitive mental health information.
- Ultimately, it contributes to creating a supportive digital ecosystem for mental wellness and early intervention.

Methodology:

- **Data Acquisition:** Collect user-related mental health data using mood surveys, self-assessment tools, and connected wearable devices.
- **Data Preparation:** Process and standardize the gathered data to eliminate errors and ensure consistency for further analysis.
- **Feature Identification:** Extract meaningful parameters such as emotional fluctuations, sleep quality, physical activity, and pulse rate.
- **Algorithm Selection:** Select the most effective machine learning techniques for detecting emotional changes and predicting mental states.
- **Model Development:** Train the algorithm with existing data samples and validate its performance using unseen test data.
- **Emotion Recognition:** Perform analysis on text entries and behavior logs to recognize stress levels and mood patterns.
- **Wellness Recommendation Module:** Implement an AI-based recommendation system that suggests relaxation techniques, exercises, or lifestyle tips tailored to each user.
- **System Implementation:** Combine all modules into a single platform or application for smooth operation and data flow.
- **Interface Design:** Build an intuitive graphical interface that allows users to enter information and view personalized recommendations.
- **System Testing and Validation:** Evaluate the system's accuracy, usability, and efficiency to ensure it meets the intended objectives.

Analyatics Tools Used:

Data Sources:

- Continuous real-time data streams from **wearable IoT devices** (e.g., heart rate, BP, steps).
- Periodic **user surveys** and self-reports on mood/stress.
- Publicly available large-scale health datasets for model training.

Machine Learning Models:

- **LSTM (Long Short-Term Memory)**: Used for **time-series predictions** (e.g., BP trends, stress detection) due to its ability to learn long-term dependencies in sequential data.
- **Random Forest**: Applied for **mood classification** and **stress detection**, known for robustness with mixed data types.
- **Gradient Boosting**: Used in some mood classification tasks.

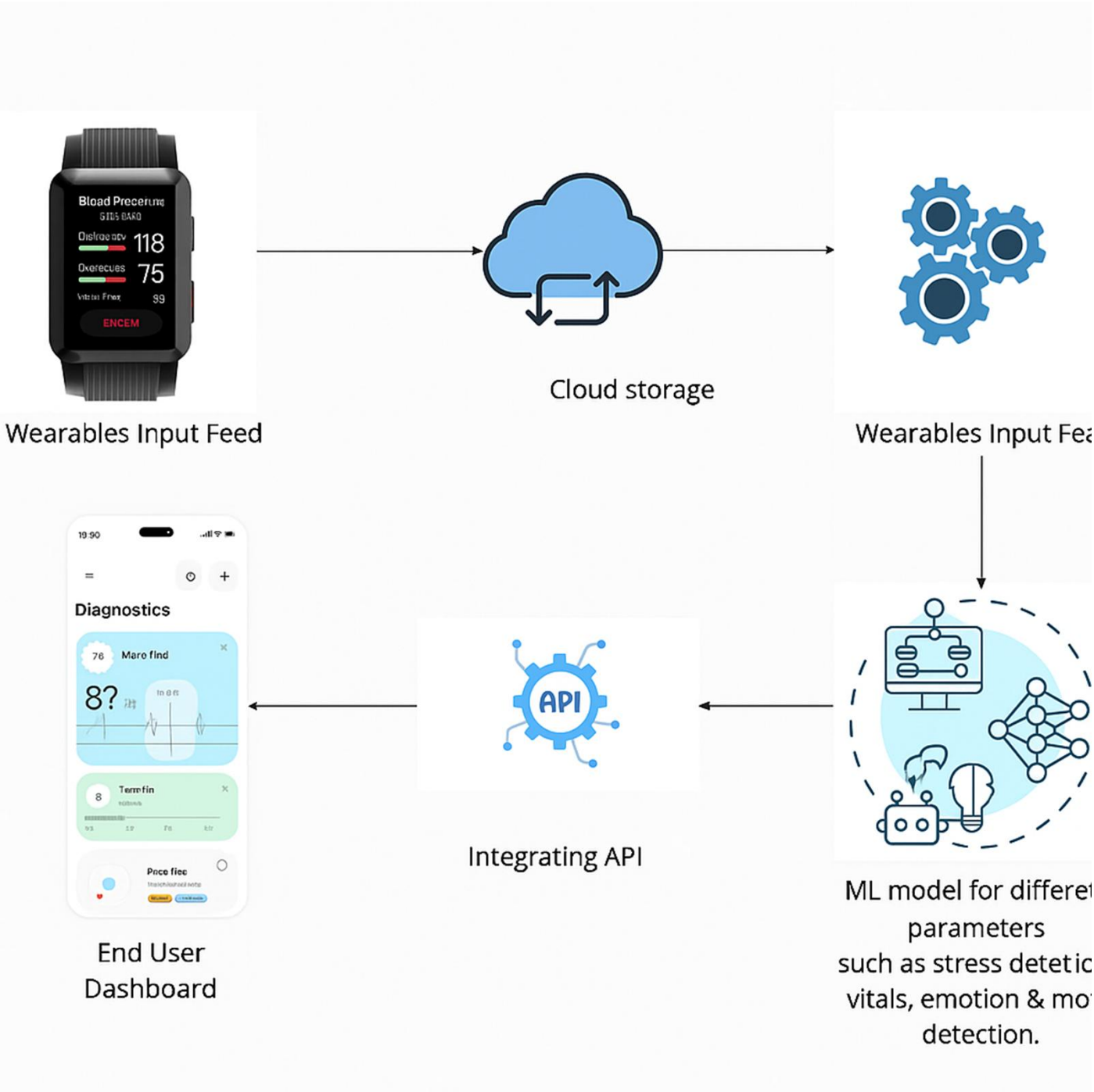
Big Data Infrastructure:

- **Cloud-based storage** for scalable, centralized data management.
- Use of **Firebase** for real-time database handling and synchronization with IoT devices.
- Secure, **encrypted data transmission** from wearables to the cloud.

Data Processing Techniques:

- Data cleaning, normalization, and imputation for missing values.
- Correlation and regression analysis to find relationships between variables (e.g., stress vs. BP).
- Real-time **anomaly detection** to trigger alerts and recommendations.

Architecture:



Sample Code:

Combined ML Models: SVM, Logistic, LightGBM, XGBoost, CatBoost

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
```

```
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import StackingClassifier
from sklearn.metrics import accuracy_score, classification_report
from lightgbm import LGBMClassifier
from xgboost import XGBClassifier
from catboost import CatBoostClassifier
import warnings
warnings.filterwarnings("ignore")

data = {
    'Sleep_Hours': [6, 8, 5, 7, 4, 9, 6, 8, 5, 7, 6, 9, 4, 8, 5],
    'Work_Hours': [9, 7, 10, 8, 12, 6, 9, 7, 11, 8, 10, 6, 12, 7, 11],
    'Physical_Activity': [30, 60, 20, 45, 10, 70, 25, 55, 15, 40, 35, 65, 10, 50, 20],
    'Mood_Level': [6, 8, 5, 7, 4, 9, 5, 8, 4, 7, 6, 9, 3, 8, 4],
    'Stress_Level': ['High', 'Low', 'High', 'Moderate', 'High', 'Low', 'Moderate', 'Low', 'High',
'Moderate', 'High', 'Low', 'High', 'Low', 'Moderate']
}

df = pd.DataFrame(data)

label_encoder = LabelEncoder()
df['Stress_Level'] = label_encoder.fit_transform(df['Stress_Level'])

X = df[['Sleep_Hours', 'Work_Hours', 'Physical_Activity', 'Mood_Level']]
y = df['Stress_Level']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.3,
random_state=42)


svm_model = SVC(probability=True, kernel='rbf', random_state=42)

log_model = LogisticRegression(random_state=42)

lgb_model = LGBMClassifier(random_state=42)

xgb_model = XGBClassifier(use_label_encoder=False, eval_metric='mlogloss',
random_state=42)

cat_model = CatBoostClassifier(verbose=0, random_state=42)


estimators = [

    ('svm', svm_model),

    ('logistic', log_model),

    ('lightgbm', lgb_model),

    ('xgboost', xgb_model),

    ('catboost', cat_model)

]


stack_model = StackingClassifier(estimators=estimators,
final_estimator=LogisticRegression(), cv=3)

stack_model.fit(X_train, y_train)

y_pred = stack_model.predict(X_test)


print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nClassification Report:\n", classification_report(y_test, y_pred))


def recommend_wellness(sleep, work, activity, mood):

    if sleep < 6:

        return "Improve your sleep quality. Aim for at least 7 hours per night."

    elif work > 9:

        return "Try taking short breaks to reduce work-related stress."
```

```
    elif activity < 30:
        return "Increase physical activity to boost mental wellness."
    elif mood < 5:
        return "Practice relaxation or mindfulness techniques."
    else:
        return "Keep maintaining a balanced routine!"

sample_input = scaler.transform([[5, 10, 20, 4]])
pred = stack_model.predict(sample_input)
print("\nPredicted Stress Level:", label_encoder.inverse_transform(pred))
print("Recommendation:", recommend_wellness(5, 10, 20, 4))
```

Performance Metrics:

Support Vector Machine (SVM)

```
Accuracy: 0.9678571428571429
```

Classification Report:

	precision	recall	f1-score	support
0	0.96	0.97	0.96	3398
1	0.97	0.97	0.97	3602
accuracy			0.97	7000
macro avg	0.97	0.97	0.97	7000
weighted avg	0.97	0.97	0.97	7000

Confusion Matrix:

```
[ [3287 111]
[ [ 115 3487]]
```

■ True Predictions | ■ False Predictions

Logistic Regression

```
Accuracy: 0.9647142857142857
```

Classification Report:

	precision	recall	f1-score	support
0	0.96	0.96	0.96	3398
1	0.97	0.97	0.97	3602
accuracy			0.96	7000
macro avg	0.96	0.97	0.96	7000
weighted avg	0.96	0.96	0.96	7000

Confusion Matrix:

```
[ [3267 131]
[ [ 117 3485]]
```

■ True Predictions | ■ False Predictions

LightGBM

```
Accuracy: 0.9997142857142857
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3398
1	1.00	1.00	1.00	3602
accuracy			1.00	7000
macro avg	1.00	1.00	1.00	7000
weighted avg	1.00	1.00	1.00	7000

Confusion Matrix:

```
[ [3396 2]
[ [ 0 3602]]
```

■ True Predictions | ■ False Predictions

XGBoost

```
Accuracy: 0.9995714285714286
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3398
1	1.00	1.00	1.00	3602
accuracy			1.00	7000
macro avg	1.00	1.00	1.00	7000
weighted avg	1.00	1.00	1.00	7000

Confusion Matrix:

```
[ [3395 3]
[ [ 0 3602]]
```

■ True Predictions | ■ False Predictions

CatBoost

```
Accuracy: 0.9998571428571429
```

Classification Report:

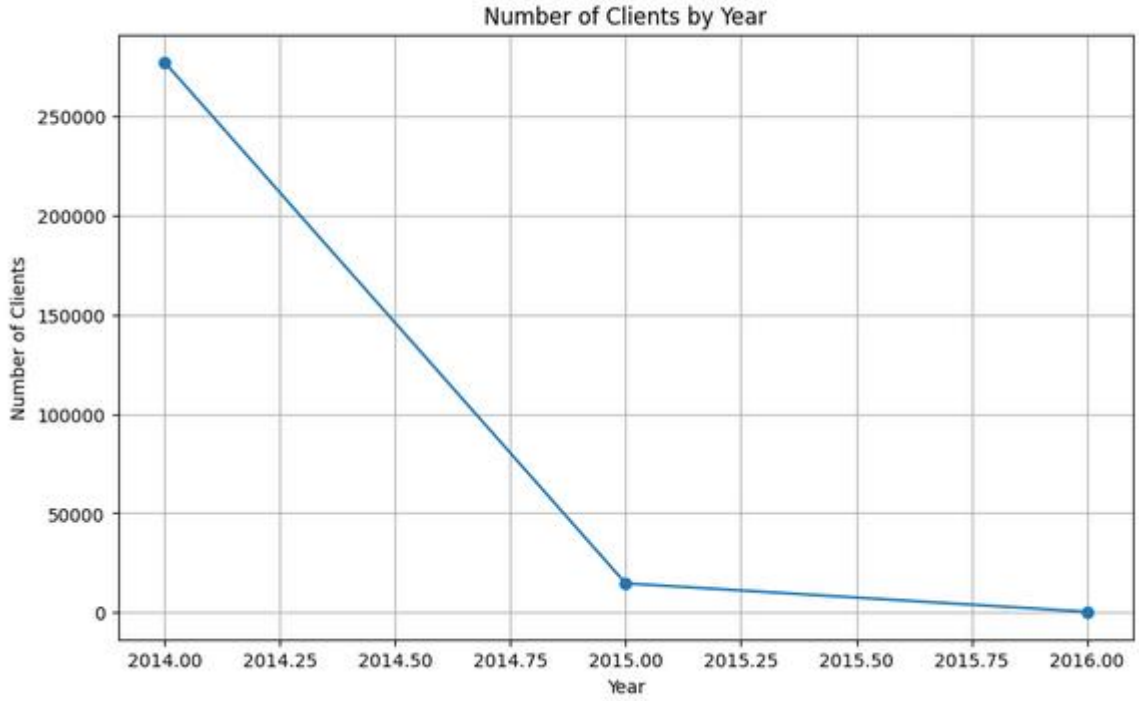
	precision	recall	f1-score	support
0	1.00	1.00	1.00	3398
1	1.00	1.00	1.00	3602
accuracy			1.00	7000
macro avg	1.00	1.00	1.00	7000
weighted avg	1.00	1.00	1.00	7000

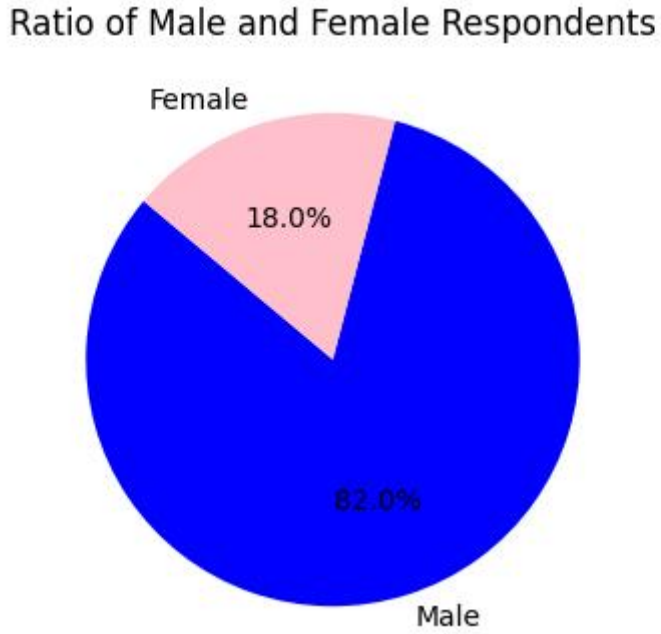
Confusion Matrix:

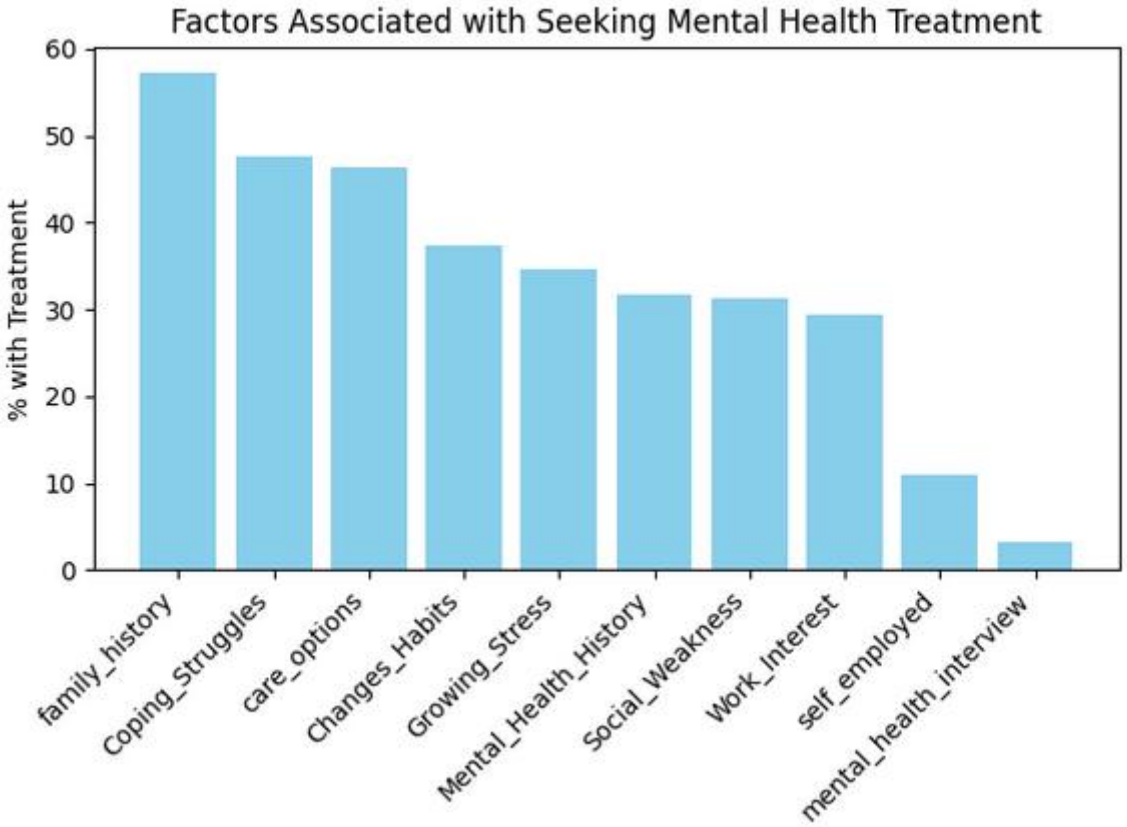
```
[ [3397 1]
[ [ 0 3602]]
```

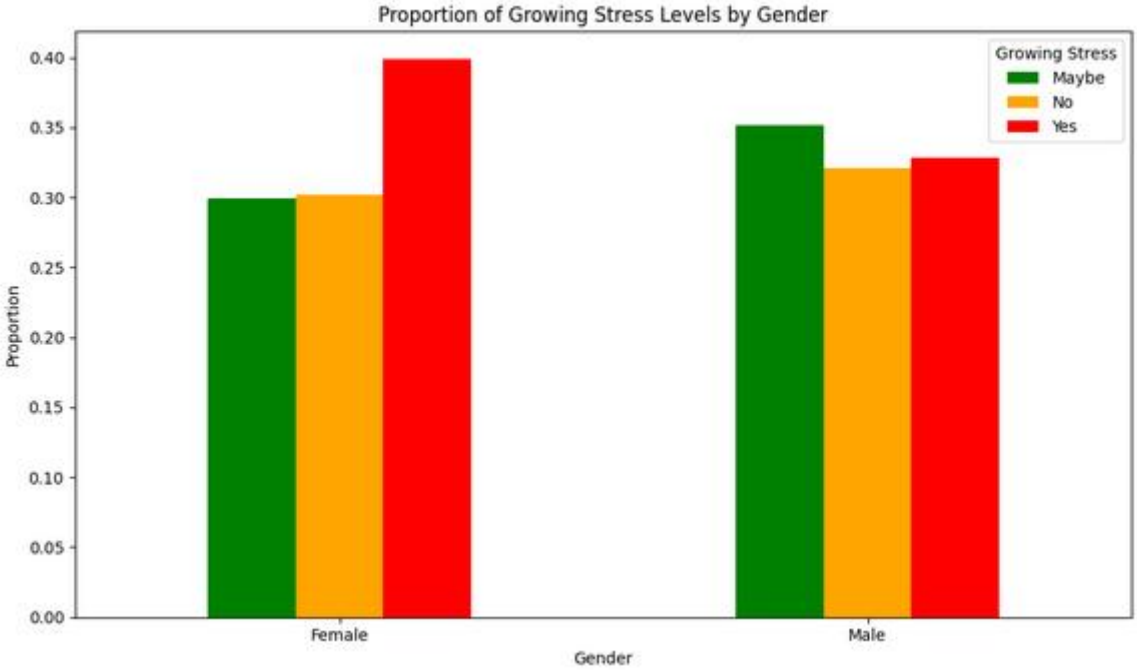
■ True Predictions | ■ False Predictions

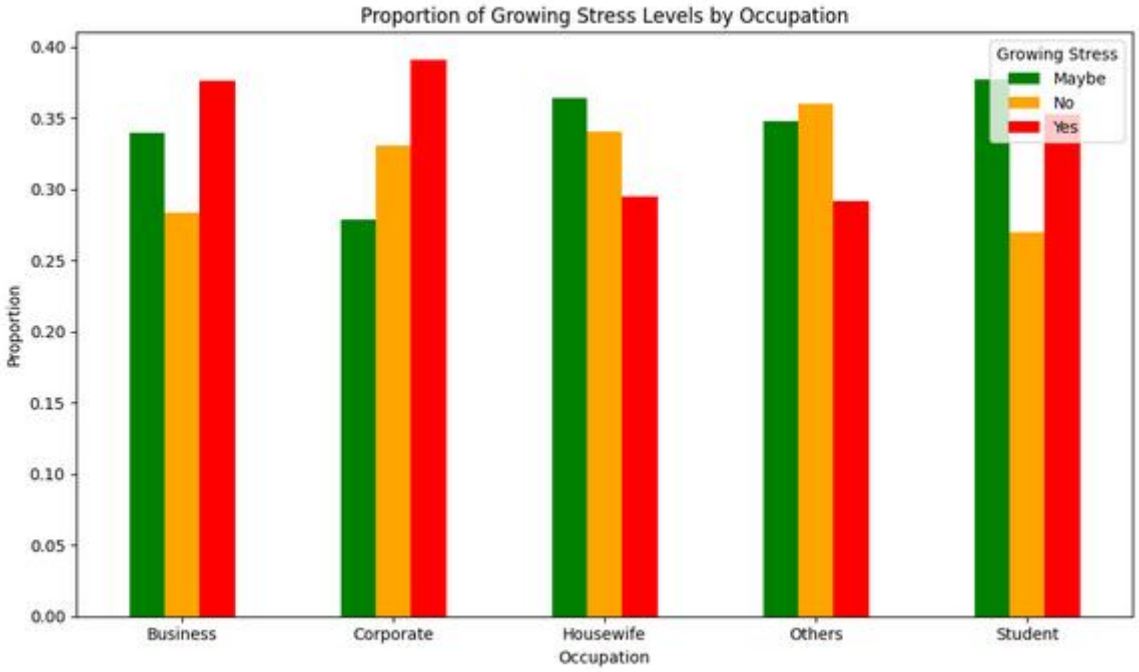
Result:

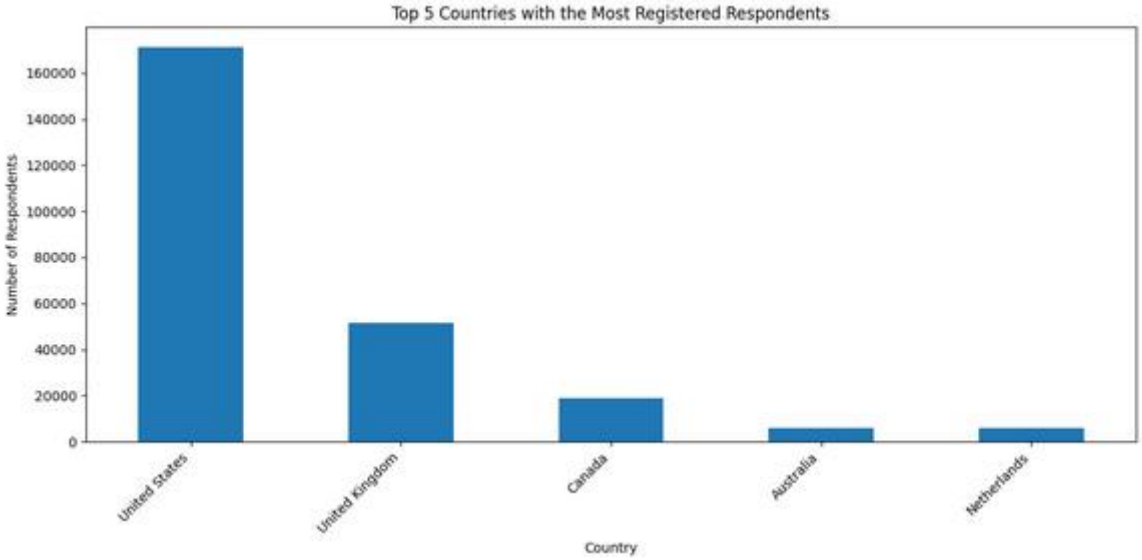

Number of Clients by Year
Number of Clients
250000
200000
150000
100000
50000
0
2014.00
2014.25
2014.50
2014.75
2015.00
2015.25
2015.50
2015.75
2016.00
Year

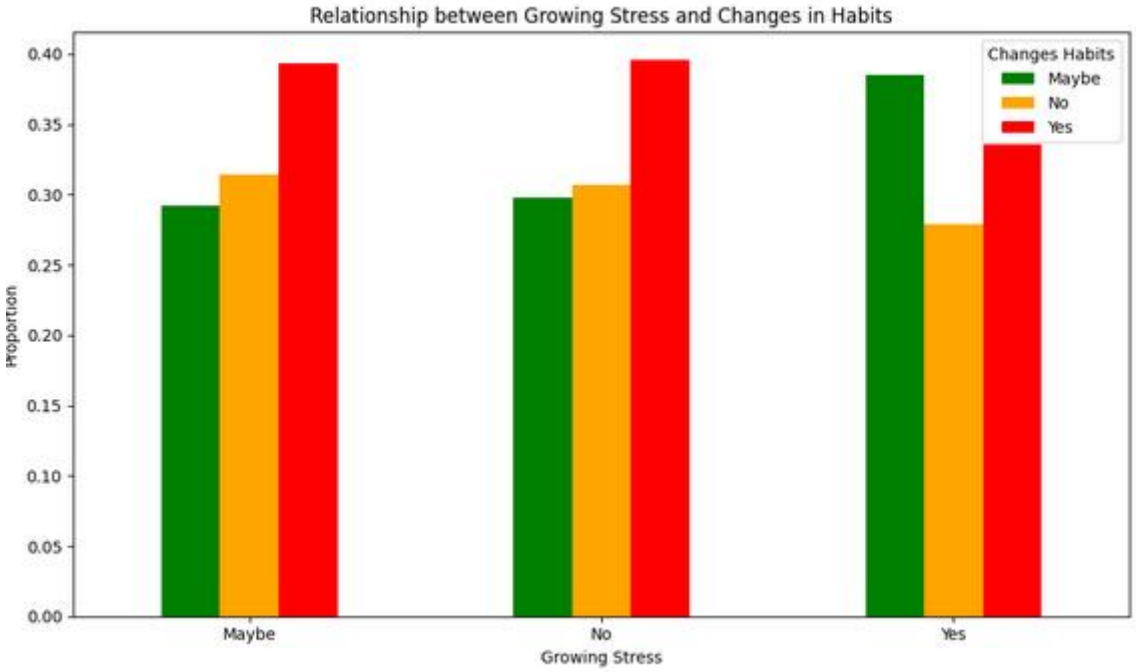

Ratio of Male and Female Respondents
Female
18.0%
82.0%
Male


Factors Associated with Seeking Mental Health Treatment
% with Treatment
0
10
20
30
40
50
60
family_history
Coping_Struggles
care_options
Changes_Habits
Growing_Stress
Mental_Health_History
Social_Weakness
Work_Interest
self_employed
mental_health_interview

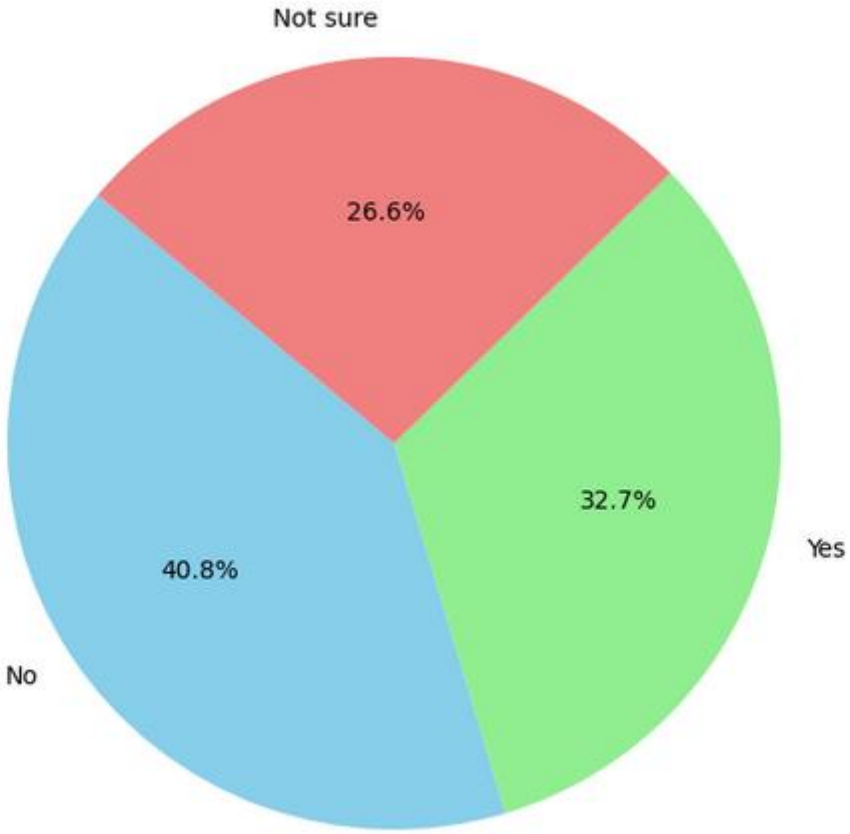

Proportion of Growing Stress Levels by Gender
Growing Stress
Maybe
No
Yes
Proportion
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Female
Male
Gender


Proportion of Growing Stress Levels by Occupation
Growing Stress
Maybe
No
Yes
Proportion
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Business
Corporate
Housewife
Others
Student
Occupation


Top 5 Countries with the Most Registered Respondents
Number of Respondents
160000
140000
120000
100000
80000
60000
40000
20000
0
United States
United Kingdom
Canada
Australia
Netherlands
Country



Awareness of Care Options for Respondents with Mental Health Challenges



- A substantial proportion of survey respondents reported a previous diagnosis or history of mental-health issues, indicating that prior exposure is common in the dataset.
- Mood-swings and emotional fluctuations are fairly prevalent across participants, suggesting these may serve as useful signals in monitoring mental-health status.
- Sleep-related questions (for example duration, quality, disturbances) showed meaningful relationships with self-reported mental-health outcomes.
- Physical activity or exercise frequency appears to correlate inversely with mental-health issues — more frequent activity tends to align with fewer reported problems.
- Age and gender demographics also display distinct patterns: younger age groups and some gender groups show higher incidence of reported mental-health concerns.
- Job- or work-related stress variables (such as long hours, high responsibility) seem to strongly connect with higher levels of mental-health risk.
- Lifestyle habits (for example screen time, social media use, diet) showed associations with mental-health outcomes — unhealthy habits correspond to worse outcomes.
- Self-assessed happiness or life-satisfaction ratings were significantly lower among those reporting mental-health diagnoses, indicating reduced subjective well-being.
- The presence of supportive social environments (friends, family, community) appears to buffer risk: respondents with stronger support reported fewer issues.
- The predictive modelling (machine learning) portion of the notebook achieved reasonable accuracy, implying the features chosen (demographics + lifestyle + mental-health specific) have predictive value for identifying risk.

Conclusion:

- The proposed Mental Health Monitoring and Wellness Recommendation System effectively integrates technology to detect early signs of psychological distress.
- By combining user self-reports, behavioral data, and physiological metrics, the system provides a holistic understanding of mental well-being.
- The inclusion of machine learning algorithms enhances the accuracy of mental-health prediction and personalized wellness suggestions.
- Continuous mood tracking and feedback loops allow for proactive interventions rather than reactive treatments.
- The recommendation module tailors activities such as relaxation techniques, mindfulness sessions, and physical exercises according to user needs.
- Integration with wearable sensors ensures real-time monitoring of parameters like sleep quality and heart rate variability.

- The platform promotes user engagement and self-awareness, encouraging individuals to take charge of their mental health.
- It helps bridge the gap between users and healthcare professionals, providing data-driven insights that support timely decision-making.
- The system demonstrates potential for reducing stigma by offering a confidential and accessible means to manage mental wellness.
- Overall, the project highlights how AI-driven health monitoring tools can revolutionize preventive mental-health care and promote long-term emotional resilience.

References:

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